

Clock Recovery

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May 26, 2026

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1 Introduction

2 Godard Timing Recovery

Timing error estimate A common timing error detector (TED) is the Gardner algorithm which computes an error term from samples $x[n]$ as [Zhou;10]

$$e[k] = \Re \{x^*[2k-1](x[2k] - x[2k-2])\} \quad (1)$$

which is then fed through a digital phase-locked loop (DPLL) and interpolator to correct timing errors. However, this requires 2 samples per symbol which is prohibitive for modern coherent receivers which operate at low oversampling rates to reduce complexity. The signal can be upsampled inside the clock recovery block to apply Gardner but a more energy-efficient approach is to use the Godard TED [Godard;78]. An RRC pulse has a bandwidth of $\frac{1+\beta}{T}$ where β is the rolloff factor and T is the symbol period, so for the Fourier transform of the input signal $X(f)$, $C(f) = X(f)X^*(f - \frac{1}{T})$ is non-zero where the spectra overlap. It is also real-valued in this region due to the symmetry of the RRC spectrum. A timing offset τ introduces a phase ramp in the frequency domain so the overlap becomes

$$C(f) = X(f)X^*(f - \frac{1}{T})e^{-j2\pi f\tau}e^{j2\pi(f-1/T)\tau} = X(f)X^*(f - \frac{1}{T})e^{-2\pi\tau/T}. \quad (2)$$

Thus the timing offset τ can be recovered from the phase of $C(f)$, which is independent of f . A discrete equivalent is given by [Han;11]

$$C[k] = X[k]X^*[k + N/2]. \quad (3)$$

The independence of the phase of $C[k]$ on k allows a more robust estimate of τ to be calculated from

$$S = \sum_{k=0}^{N/2-1} C[k]. \quad (4)$$

The standard Godard TED also appears to require 2 samples per symbol, but a modified version for arbitrary oversampling rates η is given by [Josten;17]

$$S = \sum_{k=\frac{1-\beta}{2\eta}N}^{\frac{1+\beta}{2\eta}N} X[k]X^*[k + (1 - 1/\eta)N]. \quad (5)$$

The timing error estimate normalised to the sampling period is then

$$\hat{\tau}_s = \frac{\eta \hat{\tau}}{T} = -\frac{\eta}{2\pi} \arg(S). \quad (6)$$